BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 184 · Bubble sort: trace and implement

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. For int[] a = {4,7,9}, what is a[1]?

Q2. How many times does a loop with i = 0; i < 4; i++ run?

Q3. For A=true and B=false, what is A && B?

Q4. After swapping neighbouring values 7 and 3 in [7,3,5], what is the array?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

pass — One traversal of the remaining unsorted section.

adjacent — At neighbouring indices.

swap — Exchange two stored values.

comparison — Test used to determine whether a swap is required.

early exit — Stop when a pass makes no swaps.

New learning

Compare neighbours — Compare positions j and j+1. Swap only if the earlier value is greater.

Finish one pass — The greatest unsorted value moves to the right-hand end of the unsorted section.

Repeat less work — Each completed pass fixes one final position; an entire pass with no swaps proves the array is sorted.

Worked example

Trace a complete pass of ascending bubble sort on [6,2,5,1].

1. Predict first — Compare (6,2), (6,5), then (6,1).

2. Model solution — After swaps: [2,6,5,1] → [2,5,6,1] → [2,5,1,6]. The last value is fixed.

We Do · Practise with guidance

Complete one pass on [8,3,7,4].

Predict first

Which adjacent pair is compared first?

Model solution

8 with 3, then 8 with 7, then 8 with 4. The result is [3,7,4,8].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Show one left-to-right pass on [7,3,6,2].

Question 2

Show two complete passes on [7,3,6,2].

Question 3

Why can the early-exit optimisation finish immediately on [1,2,3,4]?

Exit Ticket · Answer before checking

Bubble sort always needs to make at least one swap.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 7. C# arrays use zero-based indexing.

Q2. 4 times: i is 0,1,2,3.

Q3. false. Both must be true for AND.

Q4. [3,7,5].

You Do pages · Independent questions

Q1. [3,6,2,7]. Compare and swap 7 with 3, 6, then 2.

Q2. Pass 1 [3,6,2,7]; pass 2 [3,2,6,7].

Q3. The first pass makes no swaps. swapped remains false, so the loop breaks because the sequence is already sorted.

Exit Ticket · Model answer

False. An already sorted array needs no swaps.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: